Jump-starting verification via direct outreach

Executive Summary

Compute verification is probably the most urgent outstanding problem in AI safety today. Enabling it would allow rivals to pace or even pause the frontier, secure in the knowledge that other participants aren't sneaking ahead. In spite of that, it's critically labor-constrained: estimates are that only a few dozen people are working on it globally; for FTE it's around 25.

My theory of change is especially crisp: give academic researchers money in order to have them work on specific problems for which they are already one of the world's top experts. This tradition dates back at least to Vannevar Bush, who jump-started the Manhattan Project by coordinating directly with a handful of universities, and beginning the federal-academic partnership. The process has a historical track record, and can be incredibly fast: it has the potential to double or triple the amount of labor working on verification **before New Year's Eve**.

Matching problems to experts

Unlike many problems in AI Safety, verification is not dual-use. There's a well-defined set of 127 subproblems that need to get solved. In addition to not being dual-use, these problems don't require upskilling new entrants from scratch. On the contrary, most of these subproblems are active areas of research for various academics; it's just that those researchers are largely unaware of AI safety and the potential benefit of their work. Rather than trying to divert current scarce resources of "persons working in AI Safety" onto verification, I propose **growing the community** by reaching out to the academics directly, proactively offering funding. If, over the course of the verification work, some are especially value-aligned and choose to enter the field, so much the better.

Money moves research fast

We know this approach works: we saw it recently when "Fast Grants" were assigned during COVID. In April 2020, Tyler Cowen and team set up a fund with a 30-minute application and a promise of decisions within 48 hours; over the course of 2020 it moved more than $50 million to over 260 researchers, with terms limited to monthly one-paragraph updates and open preprints. In surveys, 64% of respondents said that the research wouldn't have happened at all without the FastGrant - that's a large counterfactual! So well-directed money can absolutely change the direction of research on extremely short timescales.

Who, and how many

What scale are we discussing? I've reviewed 52 documents that lay out the open problems (e.g. Reuel et al.'s Open Problems in Technical AI Governance, the IAPS, RAND and GovAI verification reports, the AI-2040, Amodo, Proofworks and AISF problem lists). Along

with hardware-enabled governance and chip-security literature, and the IAEA safeguards and arms-control inspection literature that verification borrows from, I've identified over 1,000 distinct questions within those 127 subproblems (the full list of subproblems, each with a source quote and current status, is at https://canaryinstitute.ai/research/open-problems/). From there I've analyzed who is already publishing results on each of those. I've identified 691 specialist academics who are already publishing on topics of vital interest to the verification push; an adversarial review confirmed 617 of them as clearly relevant (the full list of academics and the papers that bear on each problem is available on request).

The mechanism is fast and personal

Most universities have a "gift" mechanism by which research can be directed to a particular professor, and even to a particular project, with minimal overhead (often less than 10%). There are often some minor stipulations, but public-benefit work funded by a nonprofit should clear them handily. Donors typically provide a letter which specifies the faculty member and topic, but can't specify deliverables or dates. In this scenario, those are carried by the personal relationship between the funder and the specialist.

And in this proposal there *would* be a personal relationship: these requests wouldn't be coming from cold-calls to the specialist faculty. I've already created lists of all coauthors of the specialists, and from there lists of coauthors of people who are already actively working on verification. These "seed" researchers could contact the mutual coauthors and ask for a warm introduction, in which the seed could explain the details of the problem, sketch the AI Risk scenario, and offer the funding. I estimate around 330 academic specialists can be reached in this manner. 44 of our specialists are located at just four universities (Stanford, Georgia Tech, U of Florida, and U of Maryland), and 33 of those are already reachable.

The pilot: one university, ten researchers, $600k

I'm currently contacting likely researchers at those four universities to see how they respond. The pilot would go to whichever one is most receptive: roughly ten specialist faculty at that school, $60k each, for $600k in total. That's enough for each of them to, e.g., assign a graduate student to begin working on it under their supervision. Keeping the pilot on one campus means one gift office and one set of stipulations, and the researchers can talk to each other down the hall. Since money directed in this manner arrives quickly, the response time would be comparably short. Depending on results from the pilot, and funder appetite, we could have dozens of additional experts working on the issue by New Year's, and hundreds before spring.